

04 — Technical Specifications

Concrete — Raft Foundation

Grade C-40 ($f_{ck} = 40$ MPa at 28 days). Water-cement ratio max 0.45. Sulphate-resistant cement (SRPC) mandatory. Slump 100–150 mm. Independent testing every 50 m³.

Concrete — Basement Walls & Columns (B3–B1)

Grade C-35. Self-compacting concrete (SCC) for walls. Curing compound applied immediately after formwork removal.

Concrete — Superstructure Columns & Shear Walls (G–F10)

Grade C-35. High-early-strength admixture permitted subject to engineer approval.

Concrete — Superstructure Columns & Slabs (F11–F22)

Grade C-30.

Reinforcing Steel

Grade 60 deformed bars ($f_y = 420$ MPa) per ASTM A615. Yield strength verified by mill certificates. Minimum lap length 50 bar diameters.

Waterproofing — Raft & Basement Walls

Crystalline waterproofing admixture at 0.8% by cement weight PLUS external APP-modified bitumen membrane (4 mm, min.). Manufacturer's specialist to supervise application.

Formwork

Engineered formwork system capable of handling poured concrete pressure. Soffit shuttering for slabs using props at max 1.2 m centres. Slab soffit tolerances: ± 10 mm per 3 m.

Curtain Wall Glazing

Low-E double-glazed units: 6 mm Clear + 12 mm Argon + 6 mm Low-E. U-value ≤ 1.6 W/m²K. SHGC ≤ 0.35 . Aluminum frame powder-coated, thickness ≥ 2.0 mm.

Masonry (Internal Partitions)

AAC blocks (Autoclaved Aerated Concrete), density 550–650 kg/m³, 200 mm thick for primary partitions, 100 mm for secondary. Compressive strength ≥ 3.5 MPa.

Granite Flooring (Lobbies)

Imported granite minimum 18 mm thick, slip resistance R11 minimum. Laying in CM 1:3 with non-shrink grout. Expansion joints at max 4.5 m centres.

Paint

Interior walls: 2 coats premium emulsion (Dulux or equivalent). Exterior exposed concrete: elastomeric waterproof paint, minimum 200% elongation.